# 離散數學

## Homework #2 (Due date: 03/29/2022)

1. (7 points)

Determine whether these statements are true or false.

**a)** $\emptyset \in \{\emptyset\}$ **b)** $\emptyset \in \{\emptyset, \{\emptyset\}\}$
**c)** $\{\emptyset\} \in \{\emptyset\}$ **d)** $\{\emptyset\} \in \{\{\emptyset\}\}$
**e)** $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$ **f)** $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$
**g)** $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

**(a) True** **(b) True** **(c) False** **(d) True**
**(e) True** **(f) False** **(d) False**

2. (10 points)

The defining property of an ordered pair is that two ordered pairs are equal if and only if their first elements are equal and their second elements are equal. Surprisingly, instead of taking the ordered pair as a primitive concept, we can construct ordered pairs using basic notions from set theory. Show that if we define the ordered pair $(a, b)$ to be $\{\{a\}, \{a, b\}\}$, then $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$. [*Hint:* First show that $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$ if and only if $a = c$ and $b = d$.]

We must show that $\{\{a\}, \{a, b\}\} = \{\{c\}, \{c, d\}\}$ if and only if $a = c$ and $b = d$. The "if" part is immediate. So assume these two sets are equal. First, consider the case when $a \neq b$. Then $\{\{a\}, \{a, b\}\}$ contains exactly two elements, one of which contains one element. Thus, $\{\{c\}, \{c, d\}\}$ must have the same property, so $c \neq d$ and $\{c\}$ is the element containing exactly one element. Hence, $\{a\} = \{c\}$, which implies that $a = c$. Also, the two-element sets $\{a, b\}$ and $\{c, d\}$ must be equal. Because $a = c$ and $a \neq b$, it follows that $b = d$. Second, suppose that $a = b$. Then $\{\{a\}, \{a, b\}\} = \{\{a\}\}$, a set with one element. Hence, $\{\{c\}, \{c, d\}\}$ has only one element, which can happen only when $c = d$, and the set is $\{\{c\}\}$. It then follows that $a = c$ and $b = d$.

3. (6 points)

Can you conclude that $A = B$ if $A$, $B$, and $C$ are sets such that

**a)** $A \cup C = B \cup C$? **b)** $A \cap C = B \cap C$?
**c)** $A \cup C = B \cup C$ and $A \cap C = B \cap C$?

(a) We cannot conclude that A = B. For instance, if A and B are both subsets of C, then this equation will always hold, and A need not equal B.

(b) We cannot conclude that A = B; let C = ϕ, for example.

(c)

By putting the two conditions together, wecan now conclude that $A = B$. By symmetry, it suffices to prove that $A \subseteq B$. Suppose that $x \in A$. There are two cases. If $x \in C$, then $x \in A \cap C = B \cap C$, which forces $x \in B$. On the other hand, if $\notin C$, then because $x \in A \cup C = B \cup C$,we must have $x \in B$.

4. (8 points)

Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer $i$,

**a)** $A_i = \{i, i+1, i+2, \ldots\}$.

**b)** $A_i = \{0, i\}$.

**c)** $A_i = (0, i)$, that is, the set of real numbers $x$ with $0 < x < i$.

**d)** $A_i = (i, \infty)$, that is, the set of real numbers $x$ with $x > i$.

a) $A_i = \{i, i+1, i+2, \ldots\}$.

* $\bigcup_{i=1}^{\infty} A_i = \mathbb{Z}^+$; $\bigcap_{i=1}^{\infty} = \phi$, since $\cdots \subset A_2 \subset A_1$

b) $A_i = \{0, i\}$.

* $\bigcup_{i=1}^{\infty} A_i = \mathbb{N}$; $\bigcap_{i=1}^{\infty} = \{0\}$

c) $A_i = (0, i)$, that is, the set of real number $x$ with $0 < x < i$.

* $\bigcup_{i=1}^{\infty} A_i = \mathbb{R}^+$; $\bigcap_{i=1}^{\infty} A_i = (0, 1)$

d) $A_i = (i, \infty)$, that is, the set of real number $x$ with $x \leq i$.

* $\bigcup_{i=1}^{\infty} A_i = (1, \infty)$; $\bigcap_{i=1}^{\infty} = \phi$, since $\cdots \subset A_3 \subset A_2 \subset A_1 = (1, \infty)$ and $\infty$ is not a real number.

**Fuzzy sets** are used in artificial intelligence. Each element in the universal set $U$ has a **degree of membership**, which is a real number between 0 and 1 (including 0 and 1), in a fuzzy set $S$. The fuzzy set $S$ is denoted by listing the elements with their degrees of membership (elements with 0 degree of membership are not listed). For instance, we write {0.6 Alice, 0.9 Brian, 0.4 Fred, 0.1 Oscar, 0.5 Rita} for the set $F$ (of famous people) to indicate that Alice has a 0.6 degree of membership in $F$, Brian has a 0.9 degree of membership in $F$, Fred has a 0.4 degree of membership in $F$, Oscar has a 0.1 degree of membership in $F$, and Rita has a 0.5 degree of membership in $F$ (so that Brian is the most famous and Oscar is the least famous of these people). Also suppose that $R$ is the set of rich people with $R$ = {0.4 Alice, 0.8 Brian, 0.2 Fred, 0.9 Oscar, 0.7 Rita}.

5. (8 points)

The **complement** of a fuzzy set $S$ is the set $\overline{S}$, with the degree of the membership of an element in $\overline{S}$ equal to 1 minus the degree of membership of this element in $S$. Find $\overline{F}$ (the fuzzy set of people who are not famous) and $\overline{R}$ (the fuzzy set of people who are not rich).

$\overline{F}$ = {0.4 Alice, 0.1 Brian, 0.6 Fred, 0.9 Oscar, 0.5 Rita}, $\overline{R}$ = {0.6 Alice, 0.2 Brian, 0.8 Fred, 0.1 Oscar, 0.3 Rita}

6. (6 points)

The **intersection** of two fuzzy sets $S$ and $T$ is the fuzzy set $S \cap T$, where the degree of membership of an element in $S \cap T$ is the minimum of the degrees of membership of this element in $S$ and in $T$. Find the fuzzy set $F \cap R$ of rich and famous people.

{0.4 Alice, 0.8 Brian, 0.2 Fred, 0.1 Oscar, 0.5 Rita}

7. (8 points)

Find the domain and range of these functions. Note that in each case, to find the domain, determine the set of elements assigned values by the function.

**a)** the function that assigns to each nonnegative integer its last digit

**b)** the function that assigns the next largest integer to a positive integer

**c)** the function that assigns to a bit string the number of one bits in the string

**d)** the function that assigns to a bit string the number of bits in the string

a) the function that assigns to each nonnegative integer its last digit

* The domain is the set of nonnegative integers, and the range is the set of digits (0 through 9).

b) the function that assigns the next largest integer to a positive integer

* The domain is the set of positive integers, and the range is the set of integers greater than 1.

c) the function that assigns to a bit string the number of one bits in the string

* The domain is the set of all bit strings, and the range is the set of nonnegative integers.

d) the function that assigns to a bit string the number of bits in the string

* The domain is the set of all bit strings, and the range is the set of nonnegative integers (a bit string can have length 0).

8. (8 points)

Determine whether each of these functions from **Z** to **Z** is one-to-one.

**a)** $f(n) = n - 1$ **b)** $f(n) = n^2 + 1$

**c)** $f(n) = n^3$ **d)** $f(n) = \lceil n/2 \rceil$

a) $f(n) = n - 1$

* Yes, since if $n_1 - 1 = n_2 - 1$, $n_1 = n_2$

b) $f(n) = n^2 + 1$

* No, since $f(x) = f(-x)$

c) $f(n) = n^3$

* Yes, since if $n_1^3 = n_2^3$, $n_1 = n_2$

d) $f(n) = \lceil n/2 \rceil$

* No, since $f(x) = f(x+1)$ for all odd $x$s

9. (6 points)

If $f$ and $f \circ g$ are one-to-one, does it follow that $g$ is one-to-one? Justify your answer.

Suppose that $g$ is not one-to-one, there exist distince elements $a_1$ and $a_2$ such that $g(a_1) = g(a_2)$. Thus, $f(g(a_1)) = f(g(a_2))$ makes $f$ not one-to-one and we have a contradiction.

10. (12 points)

Let g: N × N → N be the function g(x, y) = (x+y)(x+y+1)/2 + y. Show that g is a bijection. Also find the inverse of g.

To show that f is bijective we need to show that f is 1-1 and onto.

[f is 1-1] We will show that if (x, y) ≠ (x', y') then f(x, y) ≠ f(x', y').

Case 1. x+y = x'+y': It follows that f(x, y) - f(x', y') = y - y' ≠ 0.

Case 2. x+y ≠ x'+y': W.l.o.g., assume x+y = x'+y' + k, for some k > 0.

Clearly, f(x, y) - f(x', y') ≠ 0

Therefore, f is 1-1.

[f is onto] We need to show that ∀ n ∈ N, ∃ x, y ∈ N such that f(x, y) = n,

Case 1. n = 1+ 2+ 3+ … +k, for some k ∈ N

f(k, 0) = k(k+1)/2 = 1+ 2+ 3+ … +k = n

Case 2. n = 1+ 2+ 3+ … +k+r, for some k ∈ N and 0 < r < k+1

f(k-r, r) = k(k+1)/2 + r = 1+ 2+ 3+ … +k+r = n

Thus f is onto.

From the above, we conclude that f is a bijection.

$$f^{-1}(n) = \begin{cases} (k,0), & n = 1 + 2 + 3 + \dots + k \\ (k-r, r), & n = 1 + 2 + 3 + \dots + k + r, 0 < r < k+1 \end{cases}$$

11. (6 points)
An employee joined a company in 2009 with a starting salary of \$50,000. Every year this employee receives a raise of \$1000 plus 5% of the salary of the previous year.

a) Set up a recurrence relation for the salary of this employee $n$ years after 2009.

* $a_n = 1 + 1.05a_{n-1}$, with $a_0 = 50$

b) What will the salary of this employee be in 2017?

* $n = 8$, $a_8$ is approximately 83.4.

c) Find an explicit formula for the salary of this employee $n$ years after 2009.

$$\begin{aligned} a_n &= 1 + 1.05a_{n-1} \\ &= 1 + 1.05(1 + 1.05a_{n-2}) \\ &= \dots \\ * \quad &= 1 + 1.05 + 1.05^2 + \cdots + 1.05^{n-1} + 1.05^n a_0 \\ &= \tfrac{1.05^n - 1}{1.05 - 1} + 50 \cdot 1.05^n \\ &= 70 \cdot 1.05^n - 20 \end{aligned}$$

12. (12 points)
Find a formula for $\sum_{k=0}^{m} \lfloor \sqrt{k} \rfloor$, when $m$ is a positive integer.
(Hint: Write down the first few terms of this sum and notice the pattern.)

If we write down the first few terms of this sum we notice a pattern. It starts $(1+1+1)+(2+2+2+2+2)+(3+3+3+3+3+3+3)+\cdots$. There are three 1's, then five 2'ss, then seven 3'ss, and so on; in general there are $(i+1)^2 - i^2 = 2i+1$ copies of $i$. So we need to sum $i(2i+1)$ for an appropriate range of values for $i$. We must find this range. It gets a little messy at the end if $m$ is such that the sequence stops before a complete range of the last value is present. Let $n = \lfloor \sqrt{m} \rfloor - 1$. Then there are $n+1$ blocks, and $(n+1)^2 - 1$ is where the next-to-last block ends. The sum of those complete blocks is $\sum_{i=1}^{n} i(2i+1) = \sum_{i=1}^{n} 2i^2 + i = n(n+1)(2n+1)/3 + n(n+1)/2$. The remaining terms in our summation all have the value $n+1$ and the number of them present is $m - ((n+1)^2 - 1)$. Our final answer is therefore $n(n+1)(2n+1)/3 + n(n+1)/2 + (n+1)(m - (n+1)^2 + 1)$.

13. (12 points)
Determine whether each of these sets is finite, countably infinite, or uncountable. For those that are countably infinite, exhibit a one-to-one correspondence between the set of positive integers and that set.

**a)** the integers greater than 10
**b)** the odd negative integers
**c)** the integers with absolute value less than 1,000,000
**d)** the real numbers between 0 and 2
**e)** the set $A \times \mathbf{Z}^+$ where $A = \{2, 3\}$
**f)** the integers that are multiples of 10

a) the integers greater than 10

* Countably infinite, $f(x) = x + 10$

b) the odd negative integers

* Countably infinite, $f(x) = 1 - 2x$

c) the integers with absolute value less than 1,000,000

* Finite, $|\{-999999, -999998, \ldots, -1, 0, 1, \ldots, 999999\}| = 1,999,999$

d) the real numbers between 0 and 2

* Uncountable.

e) the set $A \times \mathbb{Z}^+$ where $A = \{2, 3\}$

* Countably infinite, $1 \longleftrightarrow (2,1), 2 \longleftrightarrow (3,1), 3 \longleftrightarrow (2,2), 4 \longleftrightarrow (3,2), \ldots$

f) the integers that are multiples of 10

* Countably infinite, $1 \longleftrightarrow 0, 2 \longleftrightarrow 10, 3 \longleftrightarrow -10, 4 \longleftrightarrow 20, 5 \longleftrightarrow -20$

14. (6 points)
Give an example of two uncountable sets $A$ and $B$ such that $A - B$ is

**a)** finite.
**b)** countably infinite.
**c)** uncountable.

a). A = (0. 1], B = (0, 1), A – B = {1} is finite.
b). A = (0. 1) ∪ N, B = (0, 1), A – B = N is countably infinite.
c). A = (0. 2), B = (0, 1), A – B = [1, 2) is uncountable

15. (12 points)
Show that (0, 1) and R have the same cardinality.

(Hint: first show that there is a bijection between R and $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ using the tan(x) function,

then show that there is a bijection between $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ and (0, 1).)

Let $f : (-\frac{\pi}{2}, \frac{\pi}{2}) \to \mathbb{R}$ be $f(x) = tan(x)$. Then $f$ is a bijection between $(-\frac{\pi}{2}, \frac{\pi}{2})$ and $\mathbb{R}$, and $|(-\frac{\pi}{2}, \frac{\pi}{2})| = |\mathbb{R}|$. Let $g : (0, 1) \to (-\frac{\pi}{2}, \frac{\pi}{2})$ be $g(x) = \pi x - \frac{\pi}{2}$. Then $g$ is a bijection between $(0, 1)$ and $(-\frac{\pi}{2}, \frac{\pi}{2})$, and $|(0, 1)| = |(-\frac{\pi}{2}, \frac{\pi}{2})|$. Thus, $|(0, 1)| = |\mathbb{R}|$

16. (12 points)
Show that the set {(x, y)| x, y ∈Z} is countable.

List the pairs according to the value of |x| + |y|.
For pairs with the same sum of absolute values, we list these pairs according to their lexicographic order.
( (a,b) < (c,d) if a<c or a=c and b <d.)

Thus, we have the following:

(0, 0), (-1,0), (0,-1), (0,1), (1,0), (-2,0), (-1,-1), (-1,1), (0,-2), (0,2). (1.-1), (1,1), (2,0),
(-3,0), (-2,-1), (-2,1), (-1,-2), (-1,2), (0,-3), (0,3), (1,-2), (1,2), (2,-1), (2,1), (3.0), (-4,0),…
Clearly, the above is an enumeration of Z × Z, therefore Z × Z ={(x, y)| x, y ∈Z} is countable.

17. (10 points)
Let $\mathbf{A}$ be an invertible matrix. Show that $(\mathbf{A}^n)^{-1} = (\mathbf{A}^{-1})^n$ whenever $n$ is a positive integer.

We must show that $\mathbf{A}^n(\mathbf{A}^{-1})^n = \mathbf{I}$, where $\mathbf{I}$ is the $n \times n$ identity matrix. Since matrix multiplication is associative, we can write this product as

$$\mathbf{A}^n\left((\mathbf{A}^{-1})^n\right) = \mathbf{A}(\mathbf{A}\ldots(\mathbf{A}(\mathbf{A}\mathbf{A}^{-1})\mathbf{A}^{-1})\ldots\mathbf{A}^{-1})\mathbf{A}^{-1}.$$

By dropping each $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ from the center as it is obtained, this product reduces to $\mathbf{I}$. Similarly $\left((\mathbf{A}^{-1})^n\right)\mathbf{A}^n = \mathbf{I}$. Therefore by definition $(\mathbf{A}^n)^{-1} = (\mathbf{A}^{-1})^n$. (A more formal proof requires mathematical induction; see Section 5.1.)